

UKRAINIAN CATHOLIC UNIVERSITY

BACHELOR THESIS

Geospatial System for Visual Navigation

RELATED WORK SECTION

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Chapter 1

Related Work

The study of urban spatial order, street network structures, and scene uniqueness has been explored for several decades. Early research in urban planning and geography laid the groundwork for understanding how street networks shape human interaction and movement. Over time, researchers have developed methods to classify urban layouts, analyze spatial entropy, and quantify scene distinctiveness.

By looking at foundational works from the mid-20th century and comparing them with modern methodologies, we can see that while general urban morphology has been well studied, the specific topic of scene uniqueness and distinctiveness in spatial networks is an emerging area with room for further exploration.

1.1 Urban Spatial Order and Street Network Typologies

Early research on street networks can be traced back to studies in urban morphology and planning, such as by Lynch [4]. Lynch introduced the concept of imageability, where certain urban features stand out as recognizable reference points, a concept closely tied to scene uniqueness. Building on these ideas, Rémi Louf [6] developed

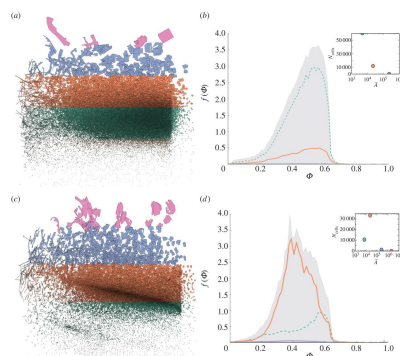


FIGURE 1.1: The fingerprints of Tokyo (a,b) and New York, NY (c,d). (a,c) from Rémi Louf [6]

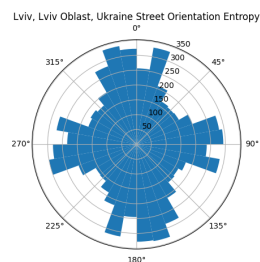


FIGURE 1.2: Street orientation entropy for Lviv, Lviv Oblast, Ukraine.

a typology of street patterns, showing how different cities exhibit unique structural characteristics. Their clustering approach classified cities into distinct groups based on the shape and distribution of urban blocks. This work provides a framework for understanding how spatial arrangements contribute to distinctiveness but does not directly focus on scene recognizability.

One of the major studies in this domain is *Urban spatial order: street network orientation, configuration, and entropy* by Boeing [2], which investigates the orientation, configuration, and entropy of street networks across 100 cities worldwide using

OpenStreetMap (OSM) [5] data and the OSMnx [1] toolkit. Boeing uses orientation entropy to measure the degrees of order and disorder in urban layouts, finding that American cities tend to be highly ordered while European cities exhibit higher entropy due to organic growth patterns - Figures 1.3 and 1.2. However, his research also does not extend to how these patterns contribute to geospatial distinctiveness from a navigational perspective.

We can see the street orientation entropy for Lviv, Lviv Oblast, Ukraine, calculated using the methodology proposed by Boeing [2] in Figure 1.2., and preliminary data for the future analysis of urban scene uniqueness in Lviv, Lviv Oblast, Ukraine and Manhattan, New York, USA in Figures 1.5 and 1.4



FIGURE 1.3: Street networks and corresponding polar histograms for Manhattan and Boston. Boeing [2]



FIGURE 1.4: Urban analysis metrics for Manhattan, New York, USA.

1.2 Quantifying Scene Uniqueness

Gudmundsson and Mohajeri [3], analyzed entropy in urban street networks. Their findings showed a connection between entropy and urban density, as well as street

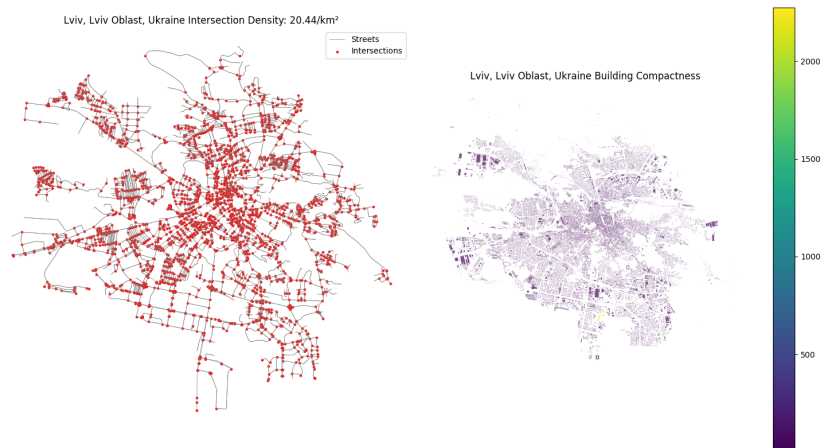


FIGURE 1.5: Urban analysis metrics for Lviv, Lviv Oblast, Ukraine.

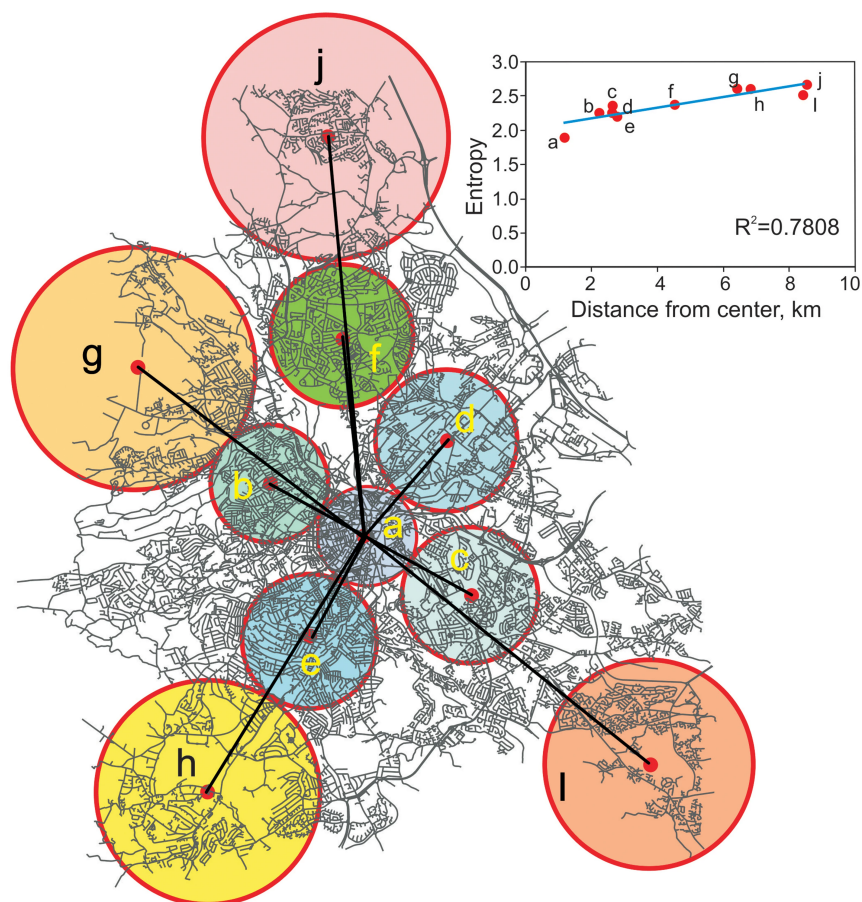


FIGURE 1.6: Variation in length entropy with distance from the centre of the street network of Sheffield. Gudmundsson and Mohajeri [3]

segment variation. These measures are useful for assessing how different urban areas develop distinct spatial characteristics. The study introduced visual representations of orientation histograms, kernel density plots, and length entropy as tools to capture the unique structure of city layouts, which are highly relevant to scene uniqueness, as well as *variation in length entropy with distance from the centre of the street network*, which can be used as a separate metric in cases when the whole area can't be viewed at once - see Figure 1.6.

1.3 Research Gaps and Contribution

Most studies analyze urban spatial order but do not explicitly state what makes a scene stand out visually or structurally. A framework for categorizing distinctive urban scenes is lacking; it is partly addressed by Gudmundsson and Mohajeri [3], but there is still a need for a systematic approach to classify and quantify scene uniqueness.

This research addresses these gaps by developing a methodology to systematically classify geospatially distinctive scenes, quantifying their distribution using entropy and street network metrics, and providing statistical analysis and visualization to understand scene uniqueness better.

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